

LINEAR MODELS

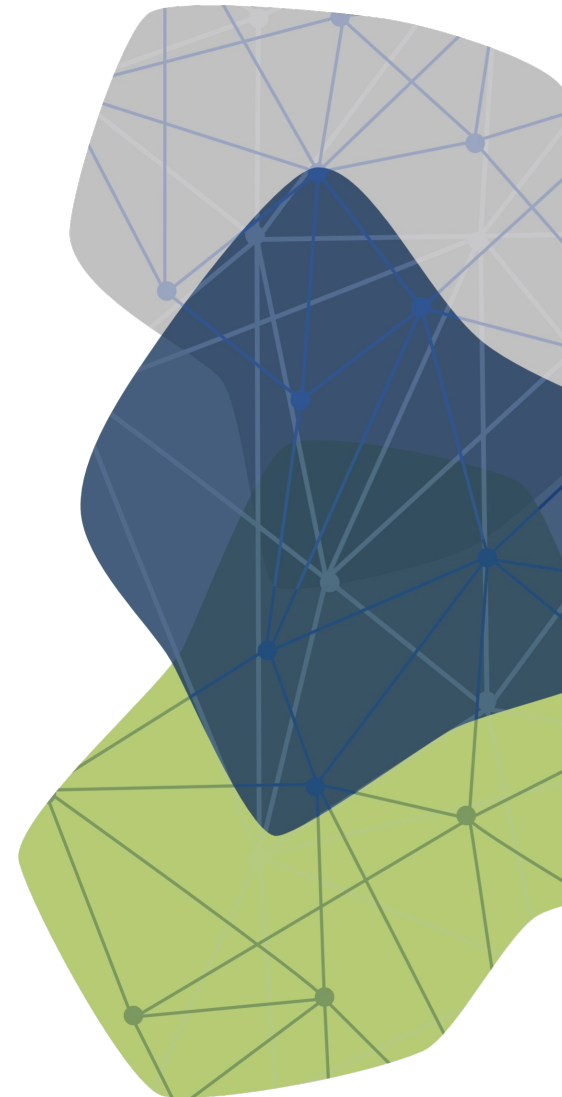
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26.05.2025



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On Iconicity

Let us begin with the analysis of Iconicity in Winter (2020: 5.1/5.2).

- The model will be relatively simplistic in only assuming two predictors: part of speech (POS) and sensory experience rating (SER).
- We restrict the data to only consider nouns and verbs.
- The data are introduced in a study by **Winter et al. (2017)**.
- While the interpretation of the categorical predictor is quite clear, of course we need to understand Iconicity **as resemblance of a sign's form to its meaning** (Winter 2020:37).
 - *“It has been proposed that sound concepts are more expressible via iconic means than concepts related to other senses, such as sight, touch, smell, or taste. This may be because auditory ideas are easier to express via imitation in an auditory medium, speech.”* (ibid.)

Revisiting Iconicity

- Let us have a look at some particularly iconic or non-iconic words, concentrating on the columns Word, POS, and Iconicity.
 - Recall that values for Iconicity can range between -5 (the word sounds the opposite of what it means) to +5 (the word sounds exactly like what it means).

```
data <- read.csv("../Data/Unit_5_BLMs/perry_winter_2017_iconicity.csv", stringsAsFactors = TRUE)

data %>% arrange(Iconicity) %>% select(Word, POS, Iconicity) %>% head()
```

##	Word	POS	Iconicity
## 1	dandelion	Noun	-2.800000
## 2	silent	Adjective	-2.166667
## 3	would	Verb	-2.100000
## 4	innocuous	Adjective	-1.923077
## 5	hamster	Noun	-1.909091
## 6	miniature	Adjective	-1.833333

Revisiting Iconicity

- Let us have a look at some particularly iconic or non-iconic words, concentrating on the columns Word, POS, and Iconicity.
 - Recall that values for Iconicity can range between -5 (the word sounds the opposite of what it means) to +5 (the word sounds exactly like what it means).

```
data %>% arrange(Iconicity) %>% select(Word, POS, Iconicity) %>% tail()
```

##		Word	POS	Iconicity
##	2996	uh oh	<NA>	4.363636
##	2997	mushy	Adjective	4.384615
##	2998	gurgle	Verb	4.416667
##	2999	click	Verb	4.461538
##	3000	hissing	Verb	4.461538
##	3001	humming	Verb	4.466667

Revisiting Iconicity

- “[S]ixty-three native English speakers rated whether a word ‘evokes a sensory experience’ on a scale from 1 to 7. ... The word with the highest sensory experience rating was garlic (6.56) ... The lowest ... rating (1.0) was shared by many words, including an, for, and hence.”
Winter et al. 2017: 8, Winter 2020: 105)
- Let us read in the data and look specifically into nouns and verbs, providing summary statistics.

```
icon <-  
  read.csv("perry_winter_2017_iconicity.csv", stringsAsFactors = TRUE) %>%  
  select(Word, Iconicity, POS, SER) %>%  
  filter(complete.cases(SER),  
         POS %in% c("Noun", "Verb")) %>%  
  droplevels() %>%  
  as.data.frame()  
  
summary(icon)
```

Revisiting Iconicity

```
icon <-  
  read.csv("perry_winter_2017_iconicity.csv", stringsAsFactors = TRUE) %>%  
  select(Word, Iconicity, POS, SER) %>%  
  filter(complete.cases(SER),  
         POS %in% c("Noun", "Verb")) %>%  
  droplevels() %>%  
  as.data.frame()
```

```
summary(icon)
```

##	Word	Iconicity	POS	SER
##	account: 1	Min. : -2.10000	Noun: 1106	Min. : 1.000
##	ache : 1	1st Qu.: 0.07143	Verb: 373	1st Qu.: 2.550
##	acid : 1	Median : 0.64286		Median : 3.270
##	add : 1	Mean : 0.80259		Mean : 3.303
##	address: 1	3rd Qu.: 1.40000		3rd Qu.: 4.000
##	adhere : 1	Max. : 4.46154		Max. : 6.560
##	(Other): 1473			

Revisiting Iconicity

Let us produce some useful summary statistics.

```
summary <-  
  icon %>%  
  group_by(POS) %>%  
  summarize(mean = mean(Iconicity, na.rm = TRUE),  
            sd = sd(Iconicity, na.rm = TRUE),  
            min = min(Iconicity, na.rm = TRUE),  
            max = max(Iconicity, na.rm = TRUE),  
            count = n()) %>%  
  as.data.frame()
```

##	POS	mean	sd	min	max	count
## 1	Noun	0.6779558	1.024389	-1.909091	4.333333	1106
## 2	Verb	1.1721430	1.192135	-2.100000	4.461538	373

Linear model 1

We will start with a model with interaction of SER and POS.

```
icon.lm <- lm(Iconicity ~ SER * POS,  
             data = icon)
```

```
##  
## Call:  
## lm(formula = Iconicity ~ SER * POS, data = icon)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -2.8448 -0.7043 -0.1257  0.5864  3.5845   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)  0.27394    0.11065   2.476  0.01341 *      
## SER          0.11817    0.03108   3.801  0.00015 ***     
## POSVerb     -0.95542    0.20971  -4.556 5.65e-06 ***     
## SER:POSVerb  0.50838    0.06535   7.780 1.36e-14 ***     
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 1.025 on 1475 degrees of freedom  
## Multiple R-squared:  0.1184, Adjusted R-squared:  0.1166   
## F-statistic: 66.05 on 3 and 1475 DF,  p-value: < 2.2e-16
```


Predictions on specified data

Since all coefficients are significant, we will extract the estimates and set up a small data frame to provide predictions on a range between 0 and 7.

```
library(broom)
```

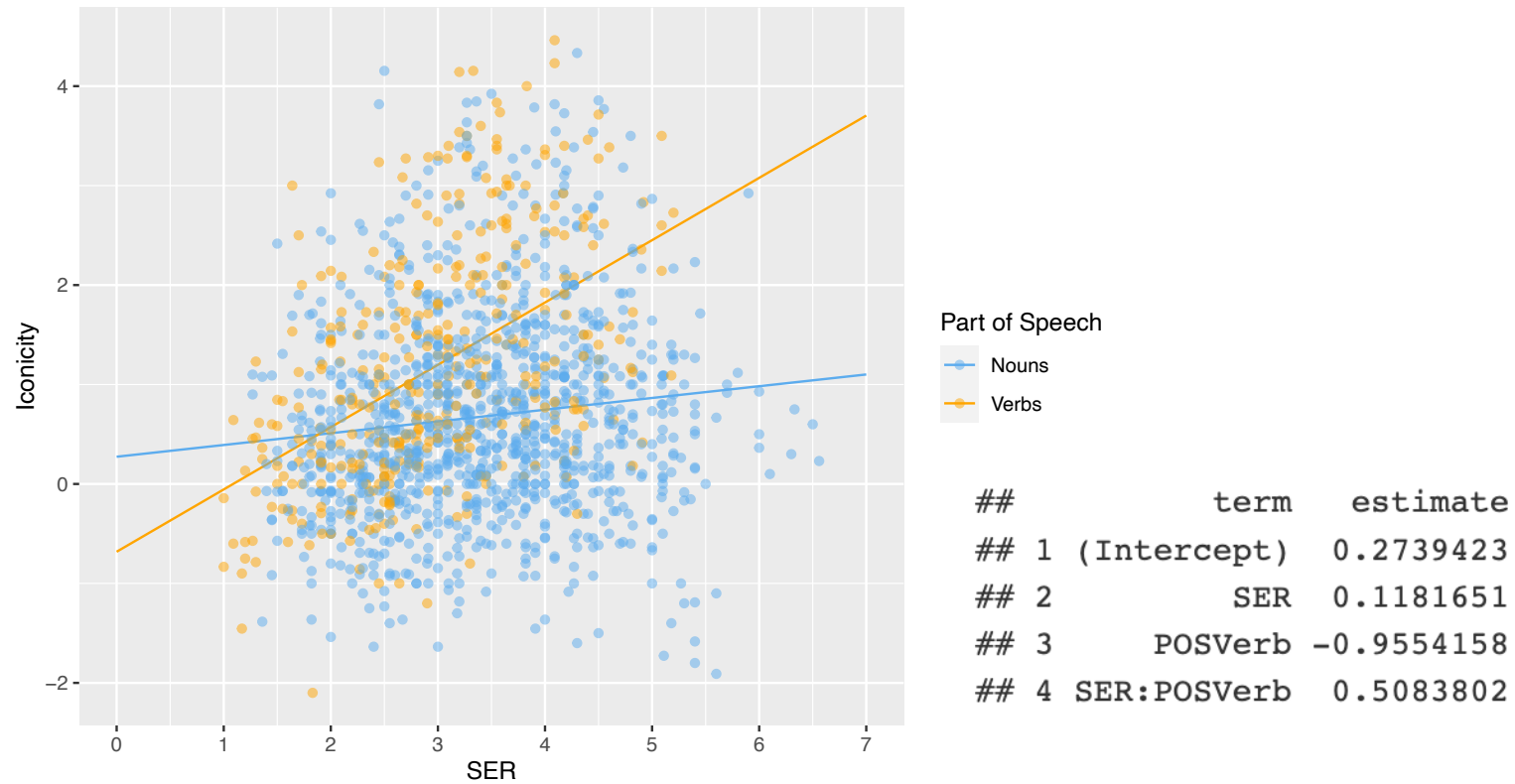
```
coefs.lm <-  
  tidy(icon.lm) %>%  
  select(term, estimate) %>%  
  as.data.frame()
```

##	term	estimate
## 1	(Intercept)	0.2739423
## 2	SER	0.1181651
## 3	POSVerb	-0.9554158
## 4	SER:POSVerb	0.5083802

```
predictions.df <-  
  expand.grid(POS = c("Noun", "Verb"), SER = 0:7) %>%  
  mutate(pred = predict(icon.lm, newdata = .))
```

##	POS	SER	pred
## 1	Noun	0	0.27394228
## 2	Verb	0	-0.68147351
## 3	Noun	1	0.39210737
## 4	Verb	1	-0.05492826
## 5	Noun	2	0.51027246
## 6	Verb	2	0.57161699

Predictions on specified data



On centering

- It is crucial to recall that a linear model is based on an intercept term and one or more slopes: $y \sim \beta_0 + \beta_1 x_1 \dots \beta_n x_n$. If we restrict ourselves to assuming that x_1 is numerical, and further that there is no other predictor (so $x_1 = x$), then $y \sim \beta_0 + \beta_1 x_1$ reduces to $y \sim \beta_0$ if $x = 0$. So, the intercept determines the value of y if $x = 0$, and if x and y are coordinates in a Cartesian space, the intercept provides y for $x = 0$, i.e. the position on the y -axis for $x = 0$.
- **Centering means nothing else as setting the mean to 0.** A linear model does not know what 0 means, it however calculates the intercept as the predicted (= y) value for $x = 0$.

On Centering

```
set.seed(12324)

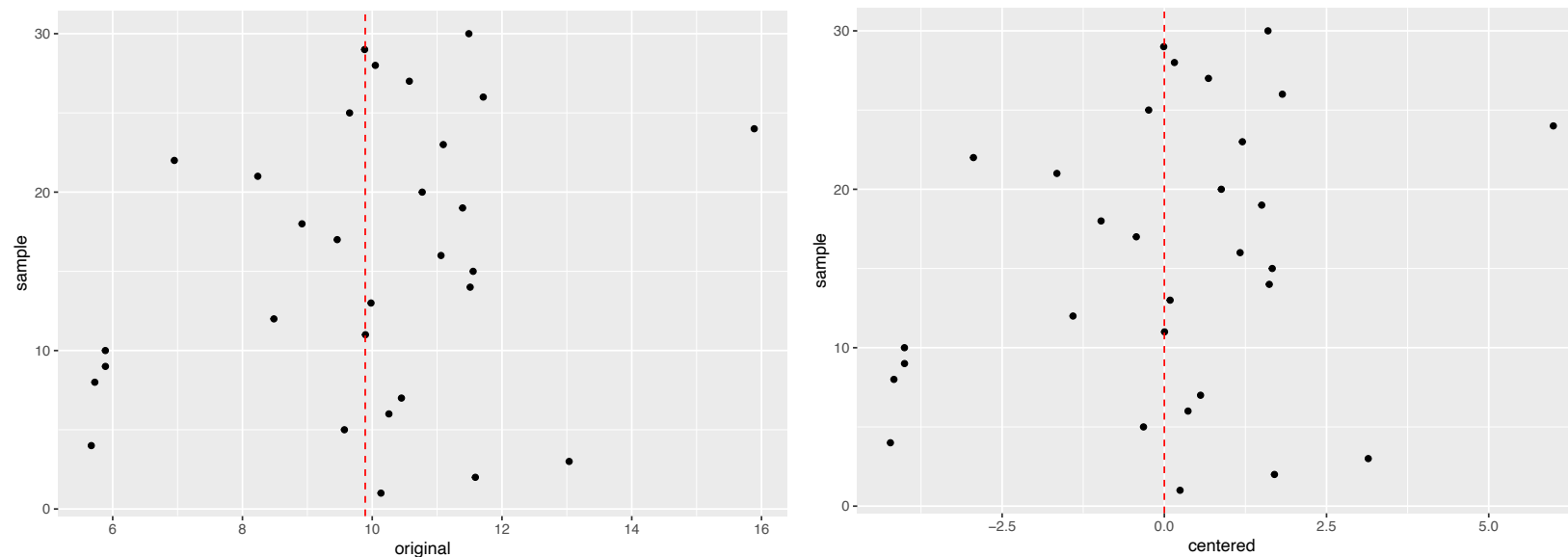
df <- data.frame(sample = 1:30, original = rnorm(n = 30, mean = 10, sd = 2))

df$centered <- df$original - mean(df$original)

summary(df)
```

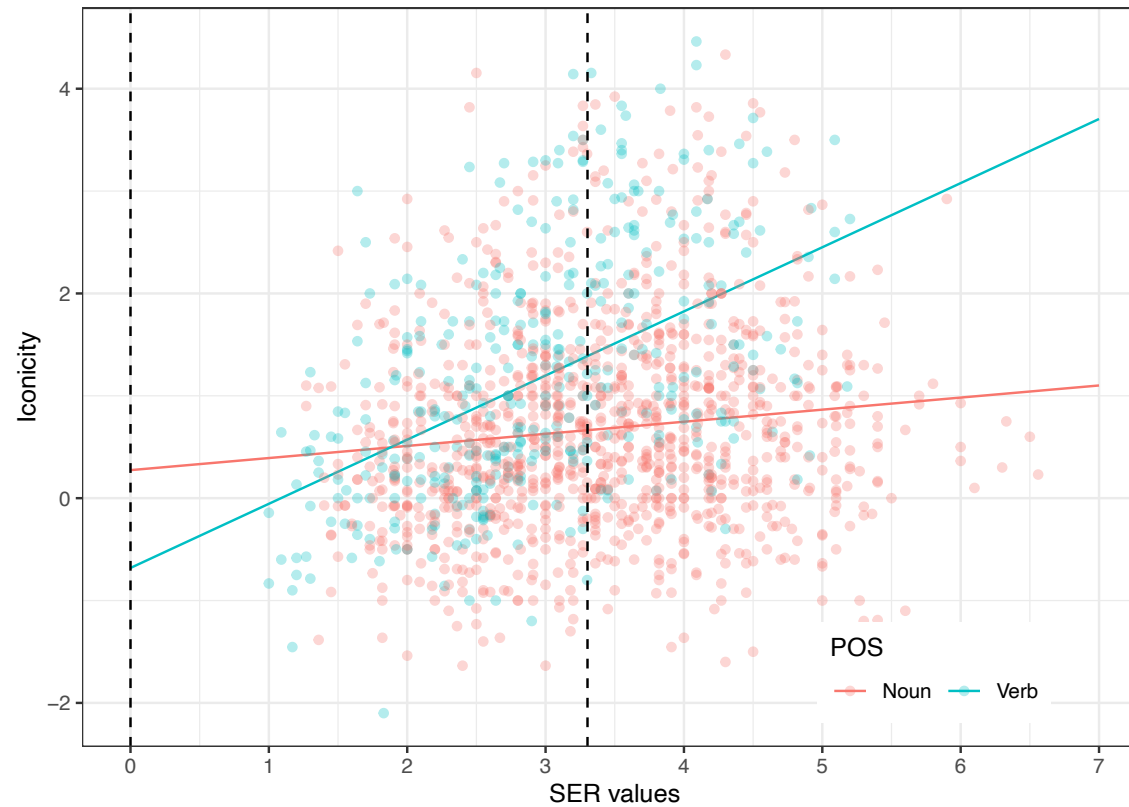
##	sample	original	centered
##	Min. : 1.00	Min. : 5.670	Min. : -4.2219
##	1st Qu.: 8.25	1st Qu.: 9.053	1st Qu.: -0.8389
##	Median :15.50	Median :10.091	Median : 0.1993
##	Mean :15.50	Mean : 9.892	Mean : 0.0000
##	3rd Qu.:22.75	3rd Qu.:11.319	3rd Qu.: 1.4266
##	Max. :30.00	Max. :15.888	Max. : 5.9959

On centering



The data points have not changed, but what has changed are the labels at the x-axis because the labels are now **centered**. In the case of our numerical predictor SER we hence get the predictors **at the mean**.

On centering



```
##           term  estimate
## 1 (Intercept)  0.2739423
## 2          SER   0.1181651
## 3      POSVerb -0.9554158
## 4 SER:POSVerb  0.5083802
```

```
##           term  estimate
## 1 (Intercept)  0.6642298
## 2      SER_c    0.1181651
## 3      POSVerb  0.7237133
## 4 SER_c:POSVerb 0.5083802
```

Multiple regression

Let us start with the data and model proposed in Winter (2020: 105ff.).

Word	POS	SER	CorteseImag
a : 1	Noun :1704	Min. :1.000	Min. :1.200
abide : 1	Verb : 557	1st Qu.:2.450	1st Qu.:3.600
able : 1	Adjective : 535	Median :3.180	Median :4.700
about : 1	Grammatical: 80	Mean :3.216	Mean :4.674
above : 1	Adverb : 39	3rd Qu.:4.000	3rd Qu.:6.100
abrasive: 1	(Other) : 32	Max. :6.560	Max. :6.900
(Other) :2995	NA's : 54	NA's :1222	NA's :1828
Conc	Syst	Freq	Iconicity
Min. :1.120	Min. :-5e-04	Min. : 1	Min. :-2.8000
1st Qu.:3.060	1st Qu.:1e-04	1st Qu.: 73	1st Qu.: 0.1538
Median :3.930	Median : 0e+00	Median : 357	Median : 0.7857
Mean :3.783	Mean : 0e+00	Mean : 11085	Mean : 0.9159
3rd Qu.:4.750	3rd Qu.: 0e+00	3rd Qu.: 1676	3rd Qu.: 1.5714
Max. :5.000	Max. : 6e-04	Max. :2134713	Max. : 4.4667
NA's :181	NA's :1898	NA's :53	
Log10Freq			
Min. :0.000			
1st Qu.:1.863			
Median :2.553			
Mean :2.596			
3rd Qu.:3.224			
Max. :6.329			
NA's :53			

Anything conspicuous about the data?

Multiple regression: Interpretation

What do the dependent and independent variables mean?

- Iconicity: *Some words sound like what they mean.* (Winter et al. 2017: 434)
- SER: *In their study, sixty-three native English speakers rated whether a word "evokes a sensory experience" on a scale from 1 to 7. ... The word with the highest sensory experience rating was **garlic** (6.56) ... The lowest ... rating (1.0) was shared by many words, including **an**, **for**, and **hence**.* (Winter et al. 2017: 435)
- CorteseImag: *"sensory experience such as a mental picture or sound"* (Winter et al. 2017: 435)
- Syst: *"each word's contribution to the overall meaning-form correlation"* (Winter et al. 2017: 435)
- Frequency: should be clear.

Multiple regression

We will include `p.value` in the tidied summary of the model, because it is important.

```
icon_mdl <- lm(Iconicity ~ SER + CorteseImag + Syst + Log10Freq, data = icon.full)
```

	term	estimate	p.value
1	(Intercept)	1.54	0.00
2	SER	0.50	0.00
3	CorteseImag	-0.26	0.00
4	Syst	401.52	0.13
5	Log10Freq	-0.25	0.00

Multiple regression: standardization

```
icon.full <-  
  icon.full %>%  
  mutate(SER_z = scale(SER),  
         CorteseImag_z = scale(CorteseImag),  
         Syst_z = scale(Syst),  
         Freq_z = scale(Log10Freq))
```

	term	estimate	p.value
1	(Intercept)	1.54	0.00
2	SER	0.50	0.00
3	CorteseImag	-0.26	0.00
4	Syst	401.52	0.13
5	Log10Freq	-0.25	0.00

	term	estimate	p.value
1	(Intercept)	1.26	0.00
2	SER_z	0.52	0.00
3	CorteseImag_z	-0.39	0.00
4	Syst_z	0.05	0.13
5	Freq_z	-0.26	0.00

Interpretation?

Multiple regression: standardization

Is it possible to determine the intercept of the standardized model from the intercept of the non-standardized one?

- What does standardization mean?
- What does it do?

	term	estimate	p.value
1	(Intercept)	1.54	0.00
2	SER	0.50	0.00
3	CorteseImag	-0.26	0.00
4	Syst	401.52	0.13
5	Log10Freq	-0.25	0.00

	term	estimate	p.value
1	(Intercept)	1.26	0.00
2	SER_z	0.52	0.00
3	CorteseImag_z	-0.39	0.00
4	Syst_z	0.05	0.13
5	Freq_z	-0.26	0.00

Multiple regression: standardization

Standardization = distance from mean / standard deviation

- `mean(SER) = 3.216549`
- `mean(CorteseImag) = 4.673828`
- `mean(Syst) = -7.991175e-06`
- `mean(Log10Freq) = 2.595575`
- `intercept_z =`
 `intercept +`
 `SER × mean(SER) +`
 `CorteseImag × mean(CorteseImag) +`
 `Syst × mean(Syst) +`
 `Log10Freq × mean(Log10Freq)`

 `= 1.25694`

Multiple regression: standardization

Assessment of standardization:

- Standardization might be a good idea if numerical predictors are on scales that differ largely, if the predictors are significant in the first place.
- This is not the case here. So we will eliminate `Syst` as a predictor.
- Before we will define a new model without `Syst`, we will check for the ranges of the other predictors to ascertain that they are on the same scale.
 - `SER`: 1.00-6.56, `CorteseImag`: 1.2-6.9, `Log10Freq`: 0.00-6.33
- **Puzzle:** What will happen if we apply `anova()` to two models including and excluding `Syst` as a predictor? Why?

	term	estimate	p.value
1	(Intercept)	1.48	0
2	<code>SER</code>	0.50	0
3	<code>CorteseImag</code>	-0.25	0
4	<code>Log10Freq</code>	-0.26	0

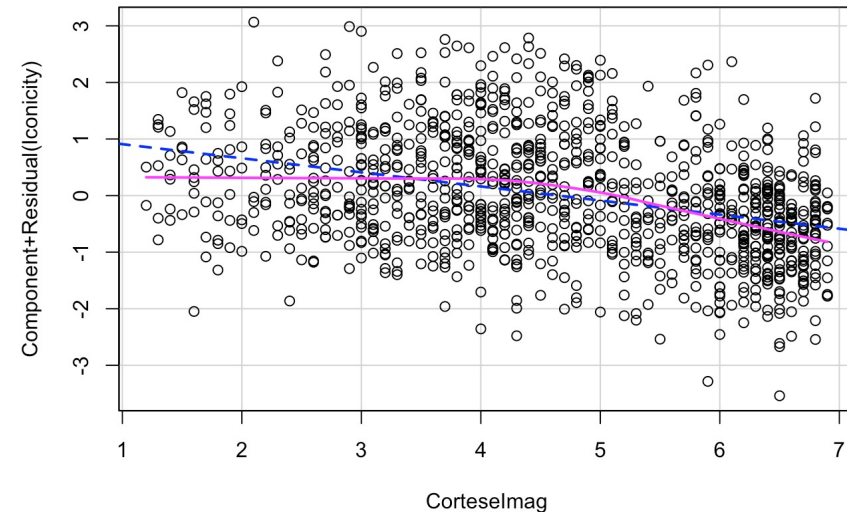
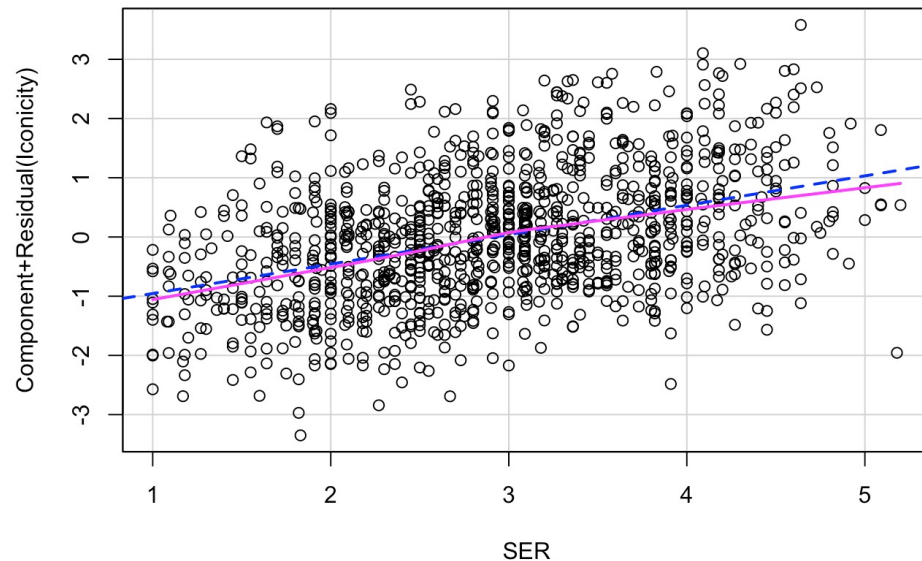
Multiple regression: Assumptions

Let us check for the regression assumptions.

- Independency of observations
- Interval-scaled response variable
- Linear relationship between quantitative independent variables and dependent variables
- Heteroscedasticity
- Multicollinearity
- Autocorrelation
- Normality of residuals

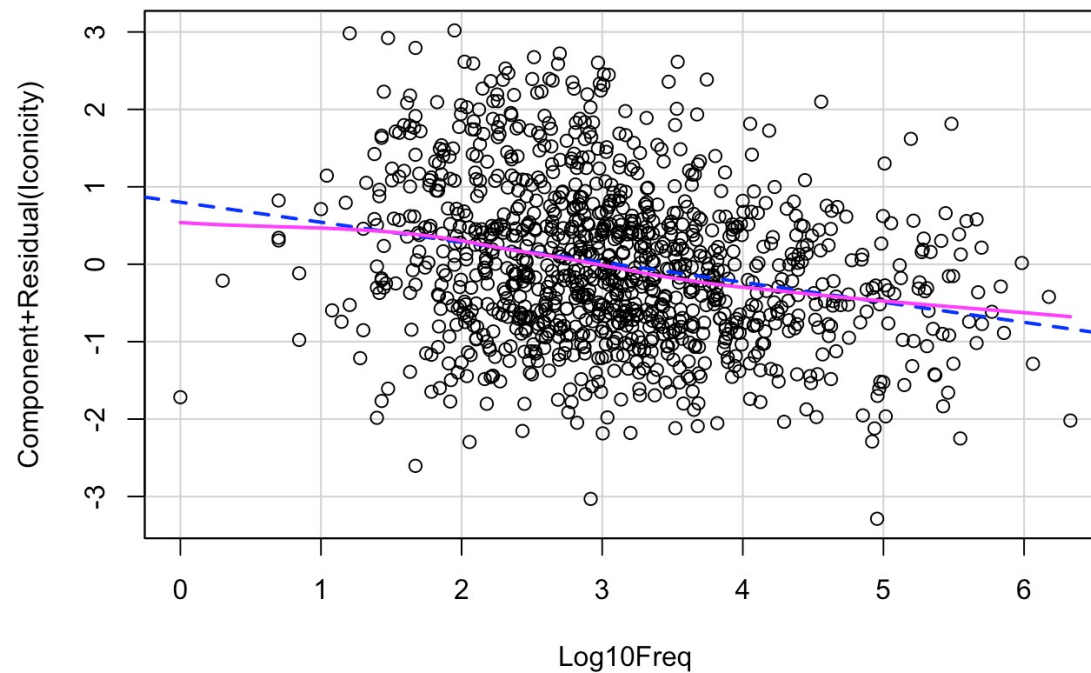
Multiple regression: Assumptions

Linear relationship between quantitative independent variables and dependent variables



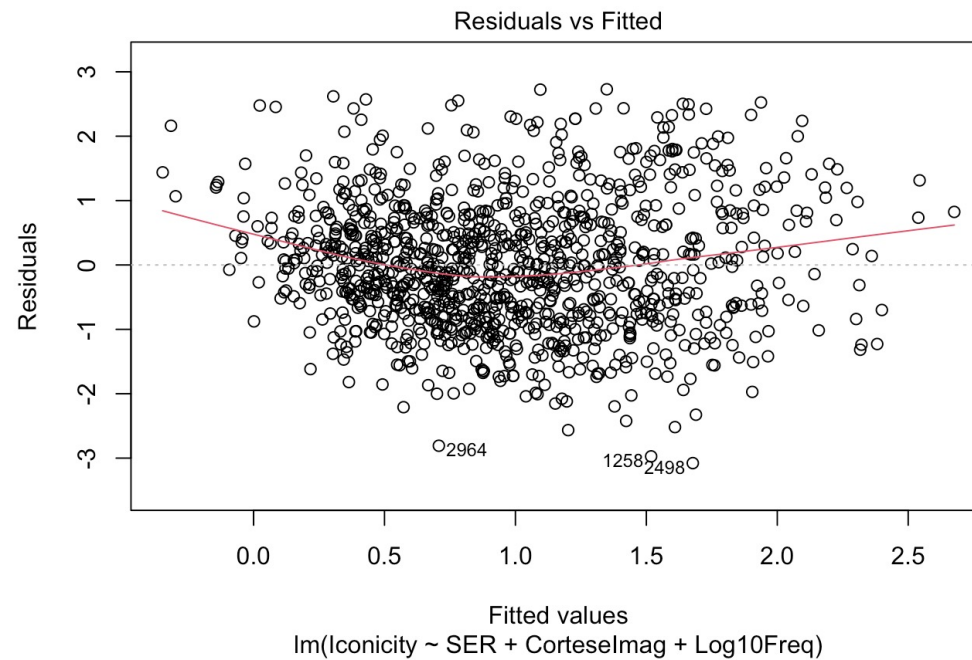
Multiple regression: Assumptions

Linear relationship between quantitative independent variables and dependent variables



Multiple regression: Assumptions

Homoscedasticity



Multiple regression: Assumptions

Homoscedasticity

- `ncvTest(icon_md1_2)`
 - Chisquare = 30.29697, Df = 1, p = 3.707e-08
- `ncvTest(icon_md1_2, ~SER)`
 - Chisquare = 4.262134, Df = 1, p = 0.038971
- `ncvTest(icon_md1_2, ~CorteseImag)`
 - Chisquare = 4.175574, Df = 1, p = 0.041011
- `ncvTest(icon_md1_2, ~Log10Freq)`
 - Chisquare = 20.96132, Df = 1, p = 4.6865e-06

Multiple regression: Assumptions

Multicollinearity

- `vif(icon_md1_2)`

	SER	CorteseImag	Log10Freq
	1.328223	1.296381	1.106594

Autocorrelation

- `durbinWatsonTest(icon_md1_2)`

lag	Autocorrelation	D-W Statistic	p-value
1	0.06731973	1.860721	0.016

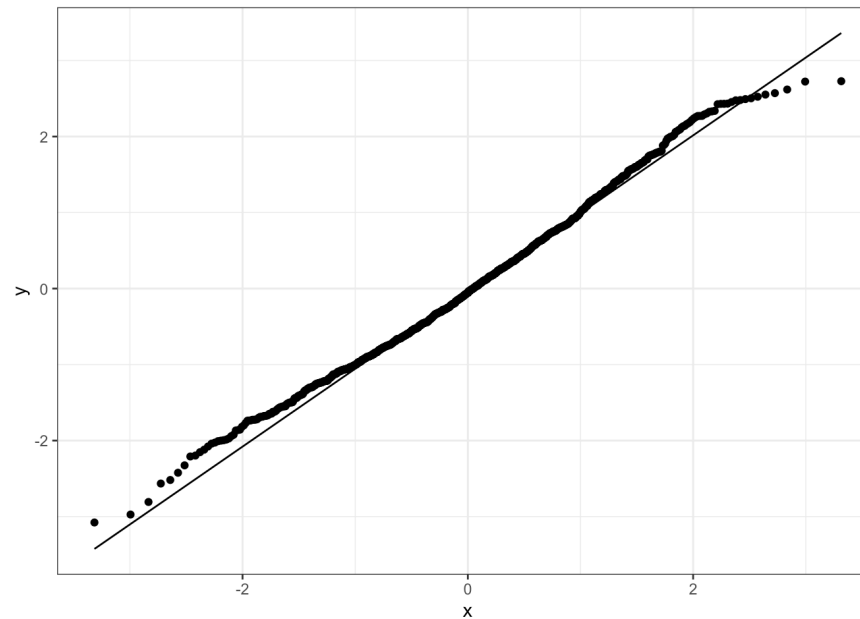
Alternative hypothesis: $\rho \neq 0$

Multiple regression: Assumptions

Normality of residuals

Shapiro-Wilk normality test

```
data:  resid.df$residuals  
W = 0.99399, p-value = 0.0002263
```



Summary: Assumptions

Several assumptions of frequentist linear models are **not** met:

- The residuals are heteroscedastic.
- The data seem to be autocorrelated.
- The distribution of the residuals diverges from a normal distribution.
- We should ask ourselves why these problems emerge. A deeper look into the data is recommended.

